Dialysis / Renal Failure

AM-03 | Adult Medical

Quick flow

01 ABCs + Dialysis History [EMT / AEMT / Paramedic]

Last completed treatment • missed/shortened sessions • access type • dry weight • residual urine

02 Hyperkalemia Suspected? [EMT / AEMT / Paramedic]

Weakness • syncope • bradycardia • hypotension • peaked T waves • PR prolongation • P-wave loss • wide QRS

03 ECG Changes / Instability? [Paramedic]

Paramedic: formulation-specific calcium dose • sodium bicarbonate up to 50 mEq • albuterol 10 mg neb when pulse present

04 Pulmonary Edema? [EMT / AEMT / Paramedic]

Upright • oxygen when indicated • early CPAP • AC-05 pathway • no routine fluid

05 Hypotension / Poor Perfusion? [AEMT / Paramedic]

Normal saline 250 mL IV/IO • reassess after each aliquot • maximum 500 mL without Medical Control

06 Access Hemorrhage? [EMT / AEMT / Paramedic]

Small gauze + focused fingertip pressure • tourniquet only for life-threatening bleeding uncontrolled by pressure

07 Infection / Peritonitis? [EMT / AEMT / Paramedic]

Fever • chills • access redness/drainage • abdominal pain • cloudy peritoneal effluent → sepsis pathway

08 Transport + Early Alert [EMT / AEMT / Paramedic]

Bring dialysis information and medication list • notify receiving facility of instability or urgent dialysis concern

Provider-Level Actions

Protect dialysis access, recognize hyperkalemia and fluid overload, and treat immediately life-threatening complications.

EMT

- Assess airway, breathing, circulation, mental status, oxygenation, perfusion, and blood glucose; provide oxygen or ventilation when clinically indicated.
- Identify the dialysis type and schedule, last completed treatment, missed or shortened sessions, access type and location, dry weight, residual urine output, medications, anticoagulants, and recent access problems.
- Protect a functioning fistula or graft extremity from blood-pressure cuffs, routine tourniquets, venipuncture, IV access, and constrictive dressings when another option exists.
- Control access hemorrhage with small gauze and focused fingertip pressure directly over the bleeding point. Request ALS early for instability, suspected hyperkalemia, pulmonary edema, or significant hemorrhage.

AEMT

- Perform all EMT actions and establish peripheral IV access away from the fistula or graft extremity when vascular treatment is indicated; use IO access when urgently required and conventional IV access cannot be obtained within the approved attempt limit.
- Apply CPAP when indicated and tolerated for pulmonary edema or respiratory failure while following the linked acute-pulmonary-edema pathway.

- For hypotension or poor perfusion without pulmonary edema, give normal saline 250 mL IV/IO and reassess lungs, oxygenation, blood pressure, and perfusion after each aliquot; maximum 500 mL without Medical Control direction.
- Do not administer calcium, sodium bicarbonate, or the high-dose hyperkalemia albuterol regimen under AM-03.

Paramedic

- Perform all EMT and AEMT actions; institute continuous cardiac monitoring and obtain a 12-lead ECG for suspected hyperkalemia, dysrhythmia, syncope, significant weakness, chest symptoms, pulmonary edema, or hemodynamic instability.
- For suspected hyperkalemia with significant ECG changes or hemodynamic instability, administer calcium and potassium-shifting treatment as specified in this protocol without delaying for laboratory confirmation.
- Treat bradycardia, cardiac arrest, pulmonary edema, shock, respiratory failure, or another concurrent emergency under the linked protocol while continuing cause-specific renal care.
- For persistent shock after cautious fluid, begin norepinephrine under the shock protocol rather than administering repeated large-volume crystalloid.

Indications

- Hemodialysis or peritoneal-dialysis patient with acute symptoms, missed treatment, or treatment complication.
- Acute or chronic renal failure with suspected hyperkalemia, pulmonary edema, shock, infection, or altered mental status.
- Bleeding, infection, malfunction, displacement, or other complication involving a dialysis fistula, graft, catheter, or peritoneal-dialysis system.

Contraindications

- Do not access or use a dialysis catheter under AM-03. Existing dialysis-catheter use during interfacility transport is governed by the separate transfer policy.
- Do not administer routine crystalloid for pulmonary edema, asymptomatic missed dialysis, or an elevated potassium concern without hypotension or poor perfusion.

Assessment | Dialysis and renal history

- Determine the cause of renal disease, dialysis type, normal schedule, last fully completed treatment, missed or shortened treatments, dry weight, recent weight change, residual urine output, and known potassium problems.
- Identify prescribed medications, potassium supplements, potassium-sparing medications, recent medication changes, anticoagulants or antiplatelet drugs, and any treatment given before EMS arrival.
- Ask about weakness, paralysis, palpitations, syncope, chest discomfort, dyspnea, orthopnea, fever, chills, nausea, vomiting, diarrhea, decreased intake, bleeding, access pain, redness, drainage, swelling, loss of thrill, pump alarms, or catheter damage.

Assessment | Suspected hyperkalemia

- Suspect hyperkalemia with renal failure, missed dialysis, known elevated potassium, crush injury, medication effect, or severe metabolic illness plus weakness, paralysis, syncope, bradycardia, hypotension, dysrhythmia, or compatible ECG changes.
- ECG findings may progress from peaked T waves to PR prolongation, diminished or absent P waves, QRS widening, bradycardia, ventricular dysrhythmia, or sine-wave morphology.
- A normal mental status or absence of classic ECG findings does not exclude dangerous hyperkalemia. Do not delay treatment of an unstable patient for laboratory confirmation.

Assessment | Dialysis-access assessment

- Identify fistula, graft, tunneled catheter, or peritoneal-dialysis access and inspect for active bleeding, aneurysmal change, trauma, erythema, warmth, tenderness, swelling, drainage, displacement, disconnection, or exposed components.
- When bleeding is controlled, assess distal circulation and note whether the usual fistula or graft thrill is present without delaying transport for hemorrhage, ischemia, or suspected thrombosis.
- In a peritoneal-dialysis patient, abdominal pain, fever, vomiting, guarding, or reported cloudy effluent should raise concern for peritonitis and sepsis.

Treatment steps

- For cardiac arrest strongly suspected to result from hyperkalemia, follow the linked arrest protocol and consider calcium and sodium bicarbonate as cause-specific treatment. Their effectiveness during hyperkalemic arrest remains uncertain under the 2025 AHA guideline.
- For pulmonary edema, position upright when tolerated, provide oxygen when indicated, initiate CPAP early when appropriate, and follow AC-05. Do not give routine fluid.
- For fistula or graft hemorrhage, use small gauze and firm focused fingertip pressure at the exact bleeding site. Avoid bulky or circumferential dressings. If immediately life-threatening extremity hemorrhage remains uncontrolled despite effective direct pressure, apply a commercial tourniquet proximal to the access, document the time, and transport emergently.
- For suspected access infection or peritoneal-dialysis peritonitis, follow the sepsis pathway. Keep peritoneal-dialysis components capped and clamped as found and do not manipulate or drain the system.
- Transport symptomatic patients and notify the receiving facility early when hyperkalemia, pulmonary edema, urgent dialysis need, uncontrolled bleeding, access infection, peritonitis, or hemodynamic instability is suspected.

Medications

Calcium gluconate 10%

1.5-3 g IV for hyperkalemia/arrhythmia; maximum 3 g per episode

- Paramedic for suspected hyperkalemia with significant ECG changes or hemodynamic instability.
- Preferred calcium formulation when administration is through a peripheral IV.
- For persistent life-threatening ECG changes, do not repeat calcium gluconate without a specific Medical Control order confirming the cumulative dose.
- Monitor the IV site continuously and stop for suspected extravasation.

Calcium chloride 10%

500-1,000 mg IV/IO for hyperkalemia/arrhythmia

- Paramedic alternative to calcium gluconate for suspected hyperkalemia with significant ECG changes or hemodynamic instability.
- Use a secure large proximal IV or IO when possible and monitor closely for extravasation.
- May repeat once after 5-10 minutes for persistent life-threatening ECG changes.

Sodium bicarbonate

1 mEq/kg IV/IO; maximum 50 mEq

- Paramedic for suspected hyperkalemia with significant ECG changes, suspected severe metabolic acidosis, or strongly suspected hyperkalemic arrest.
- Flush the line thoroughly before or after calcium administration; do not mix with calcium in the same solution or tubing.

Albuterol

10 mg by nebulizer

- Paramedic high-dose regimen for suspected hyperkalemia with a pulse.
- Do not administer nebulized albuterol during cardiac arrest.
- This dose is specific to AM-03 and is distinct from the routine bronchospasm dose.

Normal saline

250 mL IV/IO; repeat with reassessment; maximum 500 mL without Medical Control

- AEMT and Paramedic for hypotension or poor perfusion without pulmonary edema.
- Reassess lungs, oxygenation, blood pressure, and perfusion after every aliquot.
- Do not give routine fluid solely for renal failure, missed dialysis, or suspected hyperkalemia.

Warnings / precautions

- Calcium chloride and calcium gluconate are alternatives. Do not give both as the initial calcium dose.
- Calcium extravasation can cause severe tissue injury. Calcium gluconate is preferred through a peripheral IV; monitor the site continuously with either formulation.
- Do not mix calcium and sodium bicarbonate because precipitation may occur; flush the line thoroughly between administrations.
- Nebulized beta-2 agonists are not recommended as hyperkalemia treatment during cardiac arrest under the 2025 AHA guideline.
- Avoid routine or large-volume crystalloid in dialysis patients, anuric patients, pulmonary edema, or known volume overload.
- A dialysis-access hemorrhage can cause rapid exsanguination. Effective bleeding control takes priority over preservation of access patency.
- Claiborne EMS does not administer insulin for hyperkalemia under AM-03.

Clinical pearls

- Hyperkalemia may initially present with generalized weakness, paralysis, syncope, or bradycardia before a classic ECG pattern becomes obvious.
- Calcium stabilizes the myocardium but does not lower the serum potassium; definitive removal generally requires dialysis.
- Repeated large fluid boluses may worsen pulmonary edema in an anuric or volume-overloaded patient. Use cautious aliquots and early vasopressor support when shock persists.
- Small gauze and focused fingertip pressure are more effective for access bleeding than a bulky absorbent dressing that disperses pressure.
- Bring available dialysis paperwork, medication lists, recent laboratory results, and the patient's dialysis-center information with the patient.
- Document the last completed dialysis, missed time, access findings, ECG findings, medication doses, response, fluid volume, bleeding-control method, tourniquet time when applicable, and receiving-facility notification.

Special populations | Peritoneal dialysis

- Keep the transfer set and catheter capped and clamped as found. Do not manipulate, flush, disconnect, drain, or collect effluent in the field.

Special populations | Access hemorrhage

- Continue direct focused pressure during movement and transport; do not repeatedly lift the dressing to inspect the site.

Related pathways / contacts

AC-01 Asystole / PEA - Open for nonshockable cardiac arrest with suspected hyperkalemia.

AC-09 Ventricular Fibrillation / Pulseless VT - Open for shockable cardiac arrest with suspected hyperkalemia.

AC-02 Bradycardia With a Pulse - Open for symptomatic bradycardia while treating suspected hyperkalemia.

AC-05 CHF / Acute Pulmonary Edema - Open for pulmonary edema or acute decompensated heart failure.

UP-15 Suspected Sepsis - Open for suspected dialysis-access infection or peritoneal-dialysis peritonitis.

AM-05 Hypotension / Shock - Open for persistent hypotension or poor perfusion after cautious fluid.

Source and references

American Heart Association. Part 10: Adult and Pediatric Special Circumstances of Resuscitation. 2025 Guidelines for CPR and Emergency Cardiovascular Care. | National Association of State EMS Officials. National Model EMS Clinical Guidelines, Version 3, 2022. | National Kidney Foundation. Hemodialysis access care and complication guidance. | North Carolina College of Emergency Physicians. AM-03 Dialysis / Renal Failure source protocol. | Claiborne County EMS formulary and medical-director decisions, October 2026.

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